

Roll No.

TCS-704

B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

ADVANCED COMPUTER ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Elaborate the importance of instruction set architecture in computer architecture. Illustrate the five stages of an ISA/Instruction execution. (CO1)

OR

- (b) Assume a 15 cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm². Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.031 defects/cm². (CO1)

- (i) Find the yield for both wafers.

For both wafers, give your calculated value (use 3.14 for π) of die area (in cm², two decimal places) and the value of yield to four decimal places.

- (ii) Find the cost per die for both wafers.

P. T. O.

2. (a) State and prove Amdahl's laws. (CO2)

Bob is given the job to write a program that will get a speedup of 3.8 on 4 processors. He makes it 95% parallel, and goes home dreaming of a big pay raise. Using Amdahl's law, and assuming the problem size is the same as the serial version, and ignoring communication costs, what speedup will Bob actually get?

OR

- (b) What are the various performance measures available to evaluate the performance of processors? How do each of these relate to the ultimate measure of performance? (CO1)

3. (a) Critically analyse the role of Flynn classification in the history of computer evolution. (CO1)

OR

- (b) What is the cache coherence problem? Why does it occur? Explain its software and hardware based solutions in detail. (CO2)

4. (a) Discuss any five cache optimization techniques and point out the factors that get optimize. (CO2)

OR

- (b) Assume memory access to main memory on a cache "miss" takes 20 ns and memory access to the cache on a cache "hit"; takes 5 ns. If 75% of the processor's memory requests result in a cache "hit". Compute the average memory access time. (CO2)

5. (a) Consider a 16-way set-associative cache : (CO2)

- Data words are 64 bits long

(3)

- Words are addressed to the half-word
- The cache holds 2 Mbytes of data
- Each block holds 16 data words
- Physical addresses are 64 bits long

How many bits of tag, index, and offset are needed to support references to this cache ?

OR

- (b) What is virtual memory ? Why is it necessary for address translation ? Explain it with a diagram. (CO2)